Right Triangles & Trigonometry

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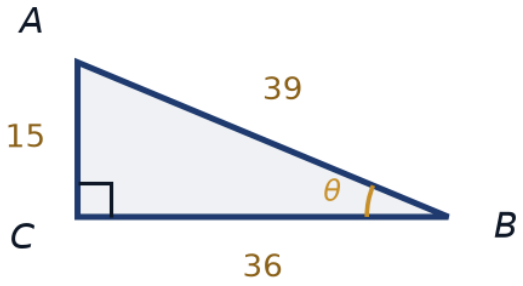
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC studying corresponding parts of congruent triangles, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 2 · MediumIn a right triangle ABC where angle $C = 90^\circ$, $\cos(A) = \frac{5}{13}$. What is the value of $\tan(A)$?

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a right triangle, acute angle θ satisfies $\sin\theta = \frac{5}{13}$. Find the exact values of $\cos\theta$, $\tan\theta$, and $\sec\theta$.

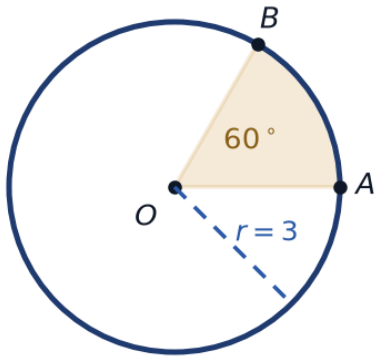
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In right triangle ABC with $C = 90^\circ$, $a = 5$, and $c = 13$. What is the value of $\sin A + \cos A$?

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 3 · AdvancedThe equation of a circle in the xy-plane is $x^2 - 8x + y^2 + 6y = 24$. What are the coordinates of the center and the radius of the circle?



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 4 · SynthesisThe equation of a circle in the xy-plane is $x^2 + y^2 - 10x + 6y - 15 = 0$. What is the length of the diameter of the circle?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 1 · FoundationsA cone-shaped storage tank has a base radius of 4 feet and a height of 9 feet. What is the volume of the tank in cubic feet, rounded to the nearest tenth?

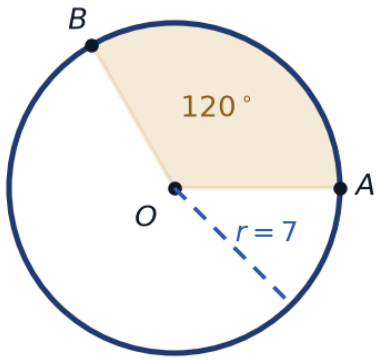
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Simplify the expression $\left(\frac{16x^8y^{-4}}{81z^{12}}\right)^{-3/4}$ and write your answer using only positive exponents.

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10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert $\frac{7\pi}{6}$ radians to degrees, and find the exact coordinates of its terminal point on the unit circle.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · AdvancedIn the system of equations below, k is a constant. For what value of k does the system have no solution? $\{kx - 6y = 15; 4x - 3y = 9$

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A right triangle has opposite 3 and hypotenuse 5. Find $\sin\theta$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Evaluate the complex expression $\frac{(3-2i)^2}{1+i}$ and write the result in standard form $a + bi$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In triangle XYZ, $XY = 7$ cm, $YZ = 10$ cm, and angle $XYZ = 65^\circ$. Calculate (a) the length of XZ to 3 significant figures and (b) the area of triangle XYZ.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Foundations & Reasoning

A right triangle has $\tan\theta = \frac{3}{4}$. Find $\sin\theta$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Coordinate Geometry

If $\cos(2x + 10^\circ) = \sin(3x - 15^\circ)$, find the value of x and evaluate $\tan(x + 16^\circ)$.

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17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Probability

Which of the following expressions is equivalent to $\frac{1-\sin^2\theta}{\cos\theta}$ for all values of θ where the expression is defined?

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Lines & Angles

If $\cos\theta = -\frac{4}{5}$ and θ lies in Quadrant II, what is the value of $\tan\theta$?

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Triangles & Congruence

If $A = (2,\ -1; 3, 4)$ and $B = (5;\ -2)$, what is the matrix product AB?

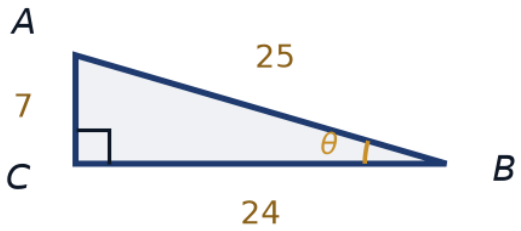
20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

Find the hypotenuse of a right triangle with legs 7 and 24.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying angles of triangles, $m\angle A = 54^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 81°

b. 91°

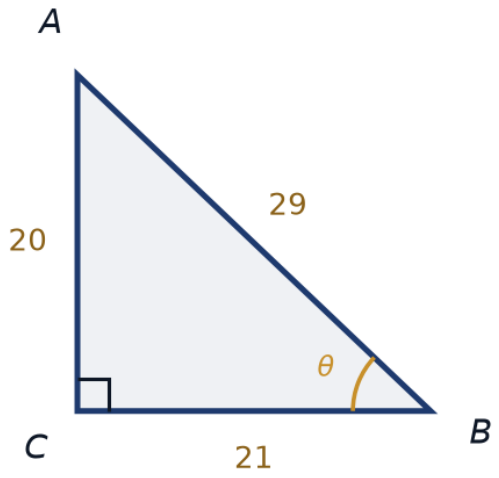
c. 73°

d. 126°

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying review: identifying congruent triangles, $m\angle A = 30^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 105°

b. 115°

c. 97°

d. 150°

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In ΔABC studying isosceles and equilateral triangles, $m\angle A = 33^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 95°

b. 105°

c. 87°

d. 147°

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In right triangle ABC, the measure of $\angle C$ is 90°. If $\sin A = \frac{x+1}{13}$ and $\cos B = \frac{3x-7}{13}$, what is the value of $\tan A$?

a. $\frac{5}{12}$

b. $\frac{12}{5}$

c. $\frac{5}{13}$

d. $\frac{12}{13}$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Right triangle legs 3 and 4: hypotenuse:

a. 3

b. 6

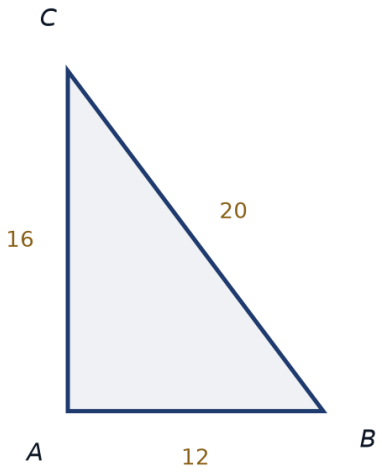
c. 7

d. 5

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A right triangle has legs 12 and 16 and hypotenuse 20. What is its area?



a. 48

b. 106

c. 96

d. 192

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A right circular cone has a base radius of 5 and a height of 12. What is the lateral area of the cone?

a. 60π

b. 65π

c. 90π

d. 130π

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right circular cylinder has a radius of 5 cm and a height of 10 cm. What is the total surface area of the cylinder?

a. 150π cm^2

b. 250π cm^2

c. 350π cm^2

d. 500π cm^2

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Two equilateral triangles have side lengths in the ratio $4:9$. What is the ratio of their altitudes?

a. $2:3$

b. $16:81$

c. $8:27$

d. $4:9$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In right ΔABC with right angle at C, altitude $CD \perp AB$. If $AD = 3$ and $DB = 12$, what is the length of leg AC?

a. $3\sqrt{5} \approx 6.71$

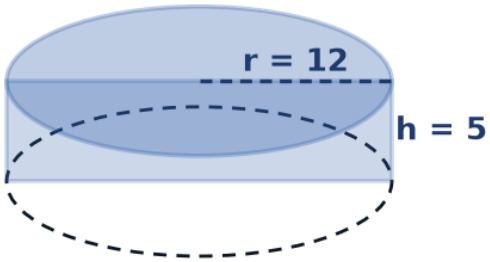
b. 6

c. 15

d. $6\sqrt{5} \approx 13.42$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A right circular cylinder has height 12 and base radius 5. A sphere is inscribed inside the cylinder such that it touches the cylinder's curved surface. What is the volume of the space inside the cylinder that is outside the sphere?



a. 100π

b. $\frac{400\pi}{3}$

c. $\frac{500\pi}{3}$

d. 200π

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In right ΔABC with right angle at C, altitude CD is drawn perpendicular to hypotenuse AB. If $AD = 4$ and $DB = 9$, what is the length of altitude CD?

a. 6.5

b. 6

c. 5.2

d. 7.2

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔJKL with right angle at vertex K, an altitude from K meets hypotenuse JL at point M. Given that segment $JM = 5$ and $ML = 15$, determine the length of leg JK.

a. $5\sqrt{3}$

b. $10\sqrt{3}$

c. 10

d. $5\sqrt{5}$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In right ΔABC with $\angle C = 90^\circ$, $AC = 6$, $BC = 8$, and hypotenuse $AB = 10$. The angle bisector of $\angle A$ intersects side BC at point D. What is the exact length of segment CD?

a. 4

b. 2.5

c. 3

d. 3.5

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In right ΔABC, altitude CD to hypotenuse AB has length 12. If segment $AD = 9$, what is the length of hypotenuse AB?

a. 16

b. 20

c. 21

d. 25

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two similar triangles have a linear scale factor of $3:7$. If the perimeter of the smaller triangle is 24, what is the perimeter of the larger triangle?

a. 48

b. 72

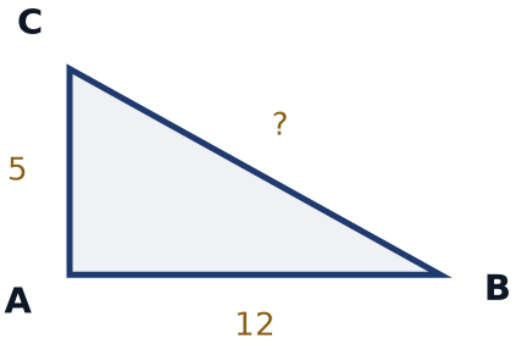
c. 130.67

d. 56

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a right triangle, the legs are 5 and 12. What is the hypotenuse?



a. 13

b. 15

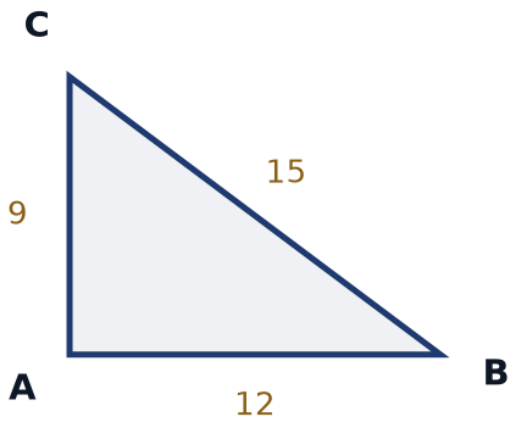
c. 17

d. $\sqrt{119}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Lines & Angles

In a right triangle, the legs have lengths 9 and 12. What is the area of the triangle?



a. 27

b. 54

c. 72

d. 108

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Triangles & Congruence

All equilateral triangles are equiangular; this triangle is equilateral, so it is equiangular. This is an example of:

a. Inductive reasoning

b. A logical fallacy

c. Circular reasoning

d. Deductive reasoning

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

All right angles measure 90°; $\angle A$ is a right angle, so $\angle A = 90^\circ$. This is an example of:

a. A logical fallacy

b. Inductive reasoning

c. Circular reasoning

d. Deductive reasoning

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A student folds several isosceles triangles in half and finds the fold is always a line of symmetry, then concludes this holds for every isosceles triangle. This is an example of:

a. Deductive reasoning

b. Inductive reasoning

c. Circular reasoning

d. A logical fallacy

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Circles

What is the contrapositive of the statement: "If Two triangles are congruent, then their corresponding sides are equal."?

a. If Their corresponding sides are equal, then two triangles are congruent.

b. If Two triangles are not congruent, then their corresponding sides are equal.

c. If Their corresponding sides are not equal, then two triangles are not congruent.

d. If Two triangles are not congruent, then their corresponding sides are not equal.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Area, Surface Area & Volume

What is the contrapositive of the statement: "If Two lines are perpendicular, then they form a right angle."?

a. If Two lines are not perpendicular, then they do not form a right angle.

b. If They do not form a right angle, then two lines are not perpendicular.

c. If Two lines are not perpendicular, then they form a right angle.

d. If They form a right angle, then two lines are perpendicular.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Transformations & Symmetry

Angles 40° and 60°: third angle:

a. 100

b. 90

c. 70

d. 80

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Foundations & Reasoning

Isosceles triangle with two sides 7 and base 10: perimeter:

a. 21

b. 17

c. 24

d. 26

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